OBED ISSAKAH

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EDUCATION

University of Nebraska–Lincoln (UNL) **Lincoln, Nebraska**
PhD, Chemical and Biomolecular Engineering *2026 – Present*
Research Focus: *Computational heterogeneous catalysis, ML/AI for materials design*

Kwame Nkrumah University of Science and Technology (KNUST) **Kumasi, Ghana**
Bachelor of Science, Materials Engineering *2019 – 2023*
First Class Honors – Cumulative Weighted Average (CWA): 73.02/100.00 (GPA: 3.7/4.0)

PUBLICATIONS AND CONFERENCES

Issakah Obed,Abdul-Manan Kayaba, Eric Kwame Anokye Asare. *Tuning of the tensile property of HDPE/CaCO3/PKS composite using machine learning.*
https://doi.org/10.1016/j.rinma.2026.100950

Daniel Nframah Ampong, Elijah Effah, Emmanuel Acheampong Tsiwah, Anuj Kumar, Emmanuel Agyekum, Esther Naa Ayorkor Doku, **Issakah Obed**, Frank Ofori Agyemang, Kwadwo Mensah-Darkwa, Ram K. Gupta. *Advances and challenges in covalent organic frameworks as an emerging class of materials for energy and environmental concerns.*
https://doi.org/10.1016/j.ccr.2024.216121

Issakah Obed, Adbul M. Kayaba, and Eric K.A Asare. *Effect of partial replacement of particulate palm kernel shell with calcium carbonate on the mechanical properties of HDPE/CaCO3 composite.*
https://doi.org/10.1016/j.rinma.2025.100668

Emmanuel Kofi Frimpong, Abdul-Manan Kayaba, Stefania Akromah1, Ezekiel Edward Nettey-Oppong, Emmanuel Essel Mensah, **Obed Issakah**, and Eric Asare. *Development and characterization of sustainable PKS/CaCO3/HDPE hybrid composites for enhanced thermal and mechanical performance.*
https://doi.org/10.1177/26349833251411841

A.M. Kayaba, **Issakah Obed**, S. Akromah, E.E. Nartey-Oppong, E.K.A Asare. *Synergistic Effects of Micro and Macro-Sized Palm Kernel Shell Fillers on the Tensile Properties of HDPE Composites.*
https://doi.org/10.1098/rsos.241911

AWARDS, GRANTS, AND HONORS

Princeton Pathways to Graduate School Scholar *August 2024*
Selected as 1 of 40 selected scholars among 250+ applicants around the world to gain graduate school application mentorship from Princeton's School of Engineering and Applied Science

Vice Chancellor's Excellence Award *August 2023*
Awarded to only 3 students in the entire university based on outstanding contribution to SDG 12 and 13.

Goldfields Ghana Scholar *September 2019-August 2023*
Awarded for high academic excellence in West Africa Certificate Examination in the community.

Responsible Artificial Intelligence Laboratory (RAIL) Grant *November 2022*
Received $1000.00 grant to for research in computational catalysis for CO2 reduction.

Materials Innovation Hub Excellence Award *August 2023*
Awarded to one person for high research involvement in a cohort of 100+ in the club.

RESEARCH EXPERIENCE

Graduate Research Assistant, Ock Lab, UNL *January 2026 – Present*
Topic: *Machine Learning for Heterogeneous Catalysis (PFAS Remediation).*
Supervisor: *Prof. Janghoon Ock, ChBE, UNL.*

- Implemented DFT workflows and machine-learned interatomic potentials (i.e. UMA) to accelerate catalyst screening for PFAS remediation, enabling rapid evaluation of adsorption/activation energetics and reaction pathways.

- Investigated governing principles of catalytic behavior by linking computed descriptors (e.g., binding energies, electronic structure metrics, surface site motifs) to activity/selectivity trends, guiding rational catalyst design.

Graduate Research Assistant, Ock Lab, UNL *January 2026 – Present*
Topic: *Graph Neural Networks for Percolation in Self-Assembled Nanomaterials.*
Supervisor: *Prof. Ock & Saraf, ChBE, UNL.*

- Modeled self-assembled nanoparticle networks as graphs and training graph neural networks (GNNs) to quantify percolation onset, conductive cluster formation, and structure–property relationships from microstructure data.
- Applied reinforcement learning (RL) to propose new network growth paths that improve percolation probability and charge-transport efficiency, supporting design of high-performance electronic materials.

Research Assistant, Computational Catalysis, KNUST *August 2022 – September 2024*
Topic: *High-throughput screening of porous materials for photoelectrochemical reduction of CO_2 into fuels using a convolutional neural network.*
Supervisor: *Prof. Luke Achenie, Virginia Polytechnic Institute and State University, USA.*

- Implemented crystal graph convolutional neural network and Monte Carlo simulation to accelerate the design of heterogeneous catalysts for carbon dioxide reduction.
- Developed interpretability models to extract physical and chemical insights from MOF structures and their catalysis behavior.

Research Assistant, KNUST *June 2025 – July 2025*
Topic: *Machine Learning for Hybrid Composite Material Design.*
Supervisor: *Dr. Eric Asare, KNUST.*

- Implemented classical ML models to predict tensile strength of HDPE/PKS/$CaCO_3$ hybrid composites using curated experimental datasets and feature engineering.
- Optimized and validated models via systematic hyperparameter tuning (grid search), reporting strong predictive accuracy (RMSE/MAE) and robust generalization using cross-validation.

Research Assistant, KNUST *August 2024 – December 2025*
Topic: *Machine learning assisted high-throughput screening for novel solid electrolyte materials discovery for solid-state sodium-ion batteries.*
Supervisor: *Prof. Luke Achenie, Virginia Polytechnic Institute and State University, USA.*

- Integrated ML models with atomistic simulations (e.g., ab initio molecular dynamics) to screen candidate solid electrolytes, prioritizing compositions with favorable Na^+ transport and stability descriptors.
- Built end-to-end workflows for model training, evaluation, and uncertainty-aware selection of promising materials to accelerate down-selection versus brute-force computation.

Research Assistant, Energy Materials and Sustainability Group, KNUST *June 2024 – December 2024*
Topic: *Development of high energy density lithium manganese oxide ($LiMn_2O_4$) batteries for tricycles.*
Supervisor: *Prof. Kwadwo Mensah Darkwa, KNUST.*

- Implemented multiphase porous electrode theory (MPET) simulator to study the open circuit voltage behavior of LMO cathode electrode at various conditions.
- Synthesizing porous microsphere $LiMn_2O_4$ cathode material for LMO batteries using solvothermal method.

Undergraduate Thesis Project, KNUST *June 2023 – August 2023*
Topic: *Effect of partial replacement of particulate palm kernel shell with calcium carbonate on the mechanical properties of HDPE/$CaCO_3$ composite.*
Supervisor: *Dr. Eric A. Asare, KNUST.*

- Developed PKS/$CaCO_3$/HDPE hybrid composite and studied the effect of adding PKS to $CaCO_3$/HDPE composite. Results showed 80% improvement in hardness (notably at 500μm PKS) and a 5% improvement in tensile strength (125μm PKS).

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TEACHING AND EXTRACURRICULAR ACTIVITIES

Teaching and Research Assistant (KNUST)*September 2023 – Present*

- Facilitated classroom discussions and assisted 100+ students in comprehending complex concepts in Physical Metallurgy and Pyrometallurgy to encourage active learning and improve class performance.
- Provided feedback on assignments and assessments, aiding students' academic growth.

Mentorship, Plastic Electronics and Photonics Student Club (KNUST)*February 2023 – Present*

- Conducted regular one-on-one sessions to address academic challenges and personal development for 7 undergraduate students in the Materials Engineering Department.

Team Lead, KNUST Green Energy Challenge Competition*May 2022 – July 2023*

- Led a team of 4 members to propose an alternative energy solution (diesel fuel extraction from plastic waste), placing in the top 3 during the penultimate stage of the competition.

SKILLS AND RELEVANT COURSES

- Python, MATLAB, Microsoft Suite
- AVOGADRO (computational chemistry), Quantum Espresso, Atomic Simulation Environment (ASE), VESTA